

SIMATS ENGINEERING

Assessment -CO3-AT1

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL,

inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

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Declaration

I, Vignesh Kumar K (Reg No: 192512375), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

Case Study 1 — Smart Medical Diagnosis System

Rules given: I. $Fever \wedge Cough \rightarrow Flu$ II. $Fever \wedge Rash \rightarrow Measles$ III. $Cough \wedge Breathlessness \rightarrow Pneumonia$.

Facts about Patient A: $Fever = True$, $Cough = True$, $Breathlessness = True$.

(a) Propositional Logic Representation

Symbol definitions:

Symbol	Meaning
F	Patient has Fever
C	Patient has Cough
R	Patient has Rash
B	Patient has Breathlessness
Fl	Patient has Flu
M	Patient has Measles
P	Patient has Pneumonia

Rules in propositional logic:

R1: $F \wedge C \rightarrow Fl$

R2: $F \wedge R \rightarrow M$

R3: $C \wedge B \rightarrow P$

Facts for Patient A: $F = True$, $C = True$, $B = True$ (R is not asserted, so it is treated as unknown/False for inference purposes.)

(b) Forward Chaining — Deriving All Diagnoses

Iter.	Rule Checked	Premise Evaluation	Result	Fact Derived
1	R1: $F \wedge C \rightarrow Fl$	$F = True, C = True \rightarrow$ premise satisfied	Fires ✓	Fl (Flu) = True
2	R2: $F \wedge R \rightarrow M$	$F = True, R = \text{unknown/False} \rightarrow$ premise NOT satisfied	Does not fire	— (Measles not derivable)
3	R3: $C \wedge B \rightarrow P$	$C = True, B = True \rightarrow$ premise satisfied	Fires ✓	P (Pneumonia) = True

No further rules can fire (all three have been checked and no new premises became satisfied). **Final derived diagnoses: Flu = True and Pneumonia = True.** Measles cannot be confirmed since Rash was never asserted.

(c) Backward Chaining — Goal Reduction Tree for 'Patient A has Pneumonia'

Goal: prove P (Pneumonia). The only rule concluding P is R3: $C \wedge B \rightarrow P$, so P reduces to the sub-goals C and B.

Goal: P (Patient A has Pneumonia)

└─ via Rule R3: $C \wedge B \rightarrow P$

└─ Sub-goal: C (Cough) $\rightarrow$ matches Fact ($C = True$) ✓ PROVED

└─ Sub-goal: B (Breathlessness) $\rightarrow$ matches Fact ($B = True$) ✓ PROVED

Both sub-goals proved as facts $\Rightarrow$ Goal P is PROVED TRUE

Conclusion: Since both sub-goals C and B are directly matched against known facts, the backward-chaining search terminates successfully, confirming '**Patient A has Pneumonia**' is TRUE.

Case Study 2 — Autonomous Traffic Management Agent

Rules:

Rule 1: $\forall x \text{ Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$
Rule 2: $\forall x \text{ ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$
Rule 3: $\forall x \forall y \text{ Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$
Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Unification of Rule 1 with the Facts

Rule 1's premise is $\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$. We unify each conjunct against the facts:

Step	Unification Performed	Result
1	UNIFY($\text{Vehicle}(x), \text{Vehicle}(\text{Ambulance})$)	$\theta_1 = \{x / \text{Ambulance}\}$
2	Apply θ_1 to remaining literal $\text{EmergencyType}(x) \rightarrow \text{EmergencyType}(\text{Ambulance})$	Substitution applied
3	UNIFY($\text{EmergencyType}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance})$)	Matches fact directly — no new bindings needed

Most General Unifier (MGU): $\theta = \{ x / \text{Ambulance} \}$

With this substitution, Rule 1 instantiates to $\text{Vehicle}(\text{Ambulance}) \wedge \text{EmergencyType}(\text{Ambulance}) \rightarrow \text{ClearPath}(\text{Ambulance})$; both premises match known facts exactly, so **ClearPath(Ambulance)** is derived.

(b) Resolution Proof that Proceed(CarA) is True

Step 1 — Convert all rules to Conjunctive Normal Form (CNF):

C1 (from Rule 1): $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$
C2 (from Rule 2a): $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$
C3 (from Rule 2b): $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$ (not needed further in this proof)
C4 (from Rule 3): $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts as unit clauses:

F1: $\text{Vehicle}(\text{Ambulance})$
F2: $\text{EmergencyType}(\text{Ambulance})$
F3: $\text{Vehicle}(\text{CarA})$
F4: $\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal (for refutation): NG: $\neg \text{Proceed}(\text{CarA})$

Step 2 — Resolution steps:

Step	Clauses Resolved	Resulting Clause
1	C1 + F1 {x/Ambulance}	$\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$
2	(Step 1) + F2	$\text{ClearPath}(\text{Ambulance})$
3	C2 + (Step 2) {x/Ambulance}	$\text{GreenSignal}(\text{Ambulance})$
4	C4 + F3 {x/CarA}	$\neg \text{Behind}(\text{CarA}, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(\text{CarA})$
5	(Step 4) + F4 {y/Ambulance}	$\neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$
6	(Step 5) + (Step 3)	$\text{Proceed}(\text{CarA})$
7	(Step 6) + NG { $\neg \text{Proceed}(\text{CarA})$ }	$\square$ (empty clause — CONTRADICTION)

The derivation of the empty clause $\square$ proves, by refutation, that the negation of the goal is unsatisfiable given the KB — therefore **Proceed(CarA) is TRUE**.

(c) Sufficiency of the KB for Two Simultaneous Emergency Vehicles

Assessment: The current KB is **NOT sufficient**. It has no mechanism to resolve conflicts if two emergency vehicles approach the same intersection at once.

Gap Identified	Why It Matters
No priority/ordering logic	If two ambulances both satisfy $\text{Vehicle}(x) \wedge \text{EmergencyType}(x)$, Rule 1 would grant ClearPath — and hence GreenSignal — to both simultaneously, with no rule deciding which should actually go first.
No conflict detection	$\text{Behind}(x, y)$ only models a single-file relationship; it cannot represent two emergency vehicles approaching from intersecting directions.
No mutual-exclusion safety constraint	Nothing in the KB prevents $\text{GreenSignal}(x)$ and $\text{GreenSignal}(y)$ from both being derived true for conflicting directions at the same time — a genuine safety gap.

Additional rules/facts needed:

- A $\text{ConflictingRoute}(x, y)$ predicate identifying when two vehicles' paths intersect.
- A $\text{Priority}(x, n)$ fact (e.g., based on arrival time, severity, or distance to the intersection) plus a rule such as: $\forall x \forall y \text{ EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{ConflictingRoute}(x, y) \wedge \text{HigherPriority}(x, y) \rightarrow \text{ClearPath}(x) \wedge \neg \text{ClearPath}(y)$.
- A mutual-exclusion safety axiom: $\forall x \forall y \text{ ConflictingRoute}(x, y) \rightarrow \neg (\text{GreenSignal}(x) \wedge \text{GreenSignal}(y))$, guaranteeing conflicting directions are never both given a green signal.

These additions are logically justified because they directly patch the identified gap — enabling the KB to correctly reason about priority and mutual exclusion — without which the existing rules could soundly derive an unsafe simultaneous ClearPath for two conflicting emergency vehicles.

Case Study 3 — Agricultural Expert Reasoning System

Knowledge Base:

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Facts: $\text{SoilDry} = \text{True}$, $\text{CropWheat} = \text{True}$

(a) Conversion to Conjunctive Normal Form (CNF)

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Step 1 (eliminate $\rightarrow$): $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

Step 2 (move $\neg$ inward): already in negation-normal form – no double negation present

Step 3 (distribute): single disjunction – already CNF

CNF result – C1: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Step 1 (eliminate $\rightarrow$): $\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Step 2 (move $\neg$ inward, De Morgan): $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

Step 3 (distribute): already a single disjunction – CNF

CNF result – C2: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Step 1 (eliminate $\rightarrow$): $\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Step 2 (move $\neg$ inward, De Morgan): $(\text{ApplyDripMethod} \vee \neg \text{CropWheat}) \vee \text{CropAtRisk}$

Step 3 (distribute/flatten): $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

CNF result – C3: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Facts as unit clauses:

C4: SoilDry

C5: CropWheat

(b) Resolution Refutation — Proving 'ApplyDripMethod'

Negated goal: NG: $\neg \text{ApplyDripMethod}$

Step	Clauses Resolved	Resulting Clause
1	C1 + C4 ($\neg \text{SoilDry} \vee \text{IrrigationNeeded}$ resolved with SoilDry)	IrrigationNeeded
2	C2 + Step1 ($\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$ resolved with IrrigationNeeded)	$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$
3	Step2 + C5 ($\neg \text{CropWheat} \vee \text{ApplyDripMethod}$ resolved with CropWheat)	ApplyDripMethod
4	Step3 + NG (ApplyDripMethod resolved with $\neg \text{ApplyDripMethod}$)	$\square$ (empty clause — CONTRADICTION)

Deriving the empty clause $\square$ proves, by refutation, that **ApplyDripMethod** is TRUE.

Resolution proof tree:

This illustrates an important reasoning pattern: removing a supporting fact from a chained rule base is an example of **non-monotonic reasoning** — unlike classical monotonic logic (where adding facts can only add conclusions, never retract them), removing a fact here retracts a whole downstream chain of derived conclusions, and the final answer for CropAtRisk genuinely depends on which reasoning convention (strict entailment vs. Closed-World Assumption) the expert system is designed to use.

Case Study 4 — Wumpus World Knowledge Agent

Rules:

R1: $\text{Stench}[x,y] \rightarrow \text{Wumpus is adjacent to } [x,y]$

R2: $\text{Breeze}[x,y] \rightarrow \text{Pit is adjacent to } [x,y]$

R3: $\text{Glitter}[x,y] \rightarrow \text{Gold is at } [x,y]$

R4: $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$

Percepts: $\text{Stench}[1,2]$, $\text{Breeze}[1,1]$, $\text{Glitter}[2,2]$ (agent currently located at $[1,1]$)

(a) Belief State Construction via Modus Ponens

Step	Percept (Premise)	Rule Instantiated	Conclusion (Modus Ponens)
1	$\text{Stench}[1,2]$	R1 instantiated at $(1,2)$	$\text{Wumpus is adjacent to } [1,2] \Rightarrow \text{Wumpus} \in \{[1,1], [1,3], [2,2]\}$
2	$\text{Breeze}[1,1]$	R2 instantiated at $(1,1)$	$\text{Pit is adjacent to } [1,1] \Rightarrow \text{Pit} \in \{[1,2], [2,1]\}$
3	$\text{Glitter}[2,2]$	R3 instantiated at $(2,2)$	Gold is at $[2,2]$

Domain knowledge refinement: since the agent is alive and currently standing at $[1,1]$, the cell it occupies cannot contain the Wumpus or a Pit (an agent cannot survive entering a cell containing either). Therefore $\neg \text{Wumpus}[1,1]$ and $\neg \text{Pit}[1,1]$ can be asserted directly, refining the Wumpus candidate set to $\{[1,3], [2,2]\}$. Applying R4 to $[1,1]$ itself ($\neg \text{Wumpus}[1,1] \wedge \neg \text{Pit}[1,1] \rightarrow \text{Safe}[1,1]$) confirms **Safe** $[1,1]$ by Modus Ponens.

All derived conclusions:

- Wumpus is located in $\{[1,3], [2,2]\}$ (refined from the initial $\{[1,1], [1,3], [2,2]\}$)
- Pit is located in $\{[1,2], [2,1]\}$
- Gold is confirmed at $[2,2]$
- Safe $[1,1]$ is confirmed (agent's current cell)

(b) Forward Chaining — Locating the Wumpus and Pit

Iter.	Action	Result
1	Start with percept facts	$\text{Stench}[1,2] = \text{True}$, $\text{Breeze}[1,1] = \text{True}$, $\text{Glitter}[2,2] = \text{True}$
2	Apply R1 to $\text{Stench}[1,2]$	$\text{Wumpus adjacent to } [1,2] \rightarrow \text{candidates } \{[1,1], [1,3], [2,2]\}$
3	Apply R2 to $\text{Breeze}[1,1]$	$\text{Pit adjacent to } [1,1] \rightarrow \text{candidates } \{[1,2], [2,1]\}$
4	Apply R3 to $\text{Glitter}[2,2]$	Gold $[2,2]$ derived
5	Apply agent-survival knowledge at $[1,1]$	$\neg \text{Wumpus}[1,1]$, $\neg \text{Pit}[1,1] \rightarrow \text{Wumpus candidates refined to } \{[1,3], [2,2]\}$
6	Apply R4 at $[1,1]$ with results of Step 5	Safe $[1,1]$ derived
7	No further rule premises are fully satisfied	Fixpoint reached — chaining stops

Conclusion: The Wumpus is in {[1,3], [2,2]} and the Pit is in {[1,2], [2,1]} — both narrowed down but not yet uniquely pinpointed with the percepts currently available.

(c) Is the Path [1,1] → [1,2] → [2,2] Safe?

Checking [1,2]:

- Wumpus check: [1,2] is not in the refined Wumpus candidate set {[1,3],[2,2]} (Rule 1 places the Wumpus adjacent to, not at, [1,2]) → $\neg \text{Wumpus}[1,2]$ can be inferred.
- Pit check: [1,2] IS one of the two Pit candidates {[1,2],[2,1]} — its status is still undetermined, so $\neg \text{Pit}[1,2]$ cannot be asserted.
- Since R4 requires BOTH $\neg \text{Wumpus}[x,y]$ and $\neg \text{Pit}[x,y]$ to conclude $\text{Safe}[x,y]$, and $\neg \text{Pit}[1,2]$ is not established, $\text{Safe}[1,2]$ CANNOT be derived — [1,2] remains a possible pit cell.

Checking [2,2]:

- Pit check: [2,2] is not in the Pit candidate set {[1,2],[2,1]} → $\neg \text{Pit}[2,2]$ can be inferred.
- Wumpus check: [2,2] IS still in the refined Wumpus candidate set {[1,3],[2,2]} — its status is undetermined, so $\neg \text{Wumpus}[2,2]$ cannot be asserted.
- Again, R4's requirement for BOTH negations is not met, so $\text{Safe}[2,2]$ CANNOT be derived — [2,2] (where the gold is) remains a possible Wumpus cell.

Conclusion: The path [1,1] → [1,2] → [2,2] **CANNOT be logically confirmed as safe** with the knowledge derived so far — the second step risks a hidden pit at [1,2], and even the destination [2,2] (with the gold) risks the Wumpus, since neither cell has been eliminated from its respective hazard candidate set. Using only Modus Ponens and Forward Chaining on the current percepts, the agent should **not** take this route; it should instead gather more percepts from a different safe cell (e.g., cautiously checking [2,1] for breeze/stench) to further narrow the Pit and Wumpus candidate sets by elimination before committing to a path toward the gold.